

**Regime Shift and the Limits of Supervised
Meta-Labeling:
Ensemble Signal Filtering on Multi-Asset
Futures**

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Abstract

This paper constructs a supervised metamodel that filters a primary directional trading signal across eleven equity, energy, and metals futures contracts using 24 years of daily data (1990–2022). The pipeline combines STFT–CVAE–GMM–HMM regime detection, 91 engineered features including TF-C self-supervised embeddings, triple-barrier labelling with four parallel volatility estimators, and an equal-weight ensemble of logistic elastic-net, XGBoost, and GRU-with-attention models. On 1,002 labelled H1 2022 test events the ensemble achieves AUC 0.509 [0.454, 0.567], not statistically separated from random. XGBoost alone reaches 0.564 [0.507, 0.624] with synthetic-null $p = 0.015$, but can only be identified as strongest by consulting the held-out set, disqualifying it under any robustness protocol. The central finding is negative: a structural inversion in the conditional direction-profitability mapping for equity index futures during the H1 2022 draw-down breaks the training-period relationship. Cluster-level feature importance confirms the inversion reproduces at a second train-test boundary, identifying it as a regime-shift signature rather than a single-split artefact. With approximately 90 effective independent observations, standard model-selection tools are unreliable, and the equal-weight composite is the assumption-minimising deliverable. A five-tier strategy construction demonstrates that the metamodel filter does not add return alpha during the regime break, while covariance-aware sizing lifts the within-filter Sharpe from 0.74 to 2.06. The pipeline diagnoses and characterises its own failure mode, establishing the appropriate boundary for supervised meta-labelling under macroeconomic regime change.

1 Introduction

Supervised meta-labelling, introduced by [López de Prado \(2018\)](#), trains a secondary classifier to decide when to trust a primary directional trading signal. The promise is that a model conditioned on volatility regime, market microstructure, and cross-asset context can filter out low-conviction trades and improve risk-adjusted returns. This paper tests that promise across eleven futures instruments spanning equity indices, energy, and metals, using a held-out test window (H1 2022) that coincides with a macroeconomic regime shift.

The contribution is primarily a negative result. The H1 2022 equity drawdown inverts the training-period relationship between signal direction and trade profitability: long signals, 75% profitable during the 2020–2021 training window, become only 31% profitable on test; shorts move from 28% to 81%. A metamodel trained on the prior regime naturally favours longs and filters them out precisely when they fail. The resulting ensemble test AUC of 0.509 is not distinguishable from random.

Three methodological commitments structure the investigation. First, the pipeline implements the [Madmoun \(2022\)](#) regime-detection spine (STFT, CVAE, GMM, HMM) end-to-end, transposed from sector-index allocation to event-level futures filtering with annual walk-forward retraining. Second, the feature set extends the regime stack with 32-dimensional TF-C self-supervised embeddings ([Zhang et al., 2022](#)), Williams swing pivots, a distress indicator, and cross-asset context features, totalling 91 columns. Third, compute is deployed as a robustness instrument: 5,000-path Politis-Romano bootstrap intervals, 200-trial TPE hyperparameter search, 30-seed neural ensembles, four parallel sigma estimators, and a 30,000-iteration WGAN-GP synthetic null, all on a single consumer GPU in 4.5 hours.

Per-event sample uniqueness following [López de Prado \(2018\)](#) AFML Chapter 4 yields approximately 90 effective independent observations, placing the analysis in the regime where estimated optimal combination weights routinely fail to beat the simple average ([Bates and Granger, 1969](#); [Claeskens et al., 2016](#); [Stock and Watson, 2004](#)). The equal-weight ensemble is therefore the assumption-minimising deliverable.

2 Data and Feature Engineering

2.1 Instruments and sample

The dataset comprises daily OHLCV histories for eleven futures contracts (four equity indices, four energy, three metals) from 1990 through June 2022. A primary directional signal $s_{i,t} \in \{-1, 0, +1\}$ is available for the 2020–2022 window. An *event* is any (date, instrument) tuple where the primary signal is non-zero; labelling every signal-

day rather than only transitions yields approximately 4,900 events. The temporal split is fixed before any modelling: training through 30 June 2021 (2,910 events), validation 1 July to 31 December 2021 (1,056 events), and test from 1 January 2022 (1,002 scored events).

2.2 Regime detection

The regime-detection spine of [Madmoun \(2022\)](#) converts each instrument’s return history into a smoothed regime posterior $p_{\text{low}} \in [0, 1]$. Four stages compose the pipeline: (i) spectral representation via Hamming-windowed DFT at 64 trading days with stride 1, producing a 16-dimensional spectral energy vector; (ii) CVAE compression into a two-dimensional latent conditioned on the sign of windowed returns; (iii) two-component GMM clustering into Low and High turbulence states, global across all eleven instruments; and (iv) HMM smoothing via the forward-backward algorithm with walk-forward refits to prevent leakage.

2.3 Feature families

The 91-column panel spans eight economic families (Table 1): 20 regime features (p_{low} , 2D latent, GMM posterior, 16 spectral bins), 32 TF-C self-supervised embeddings, 14 identity one-hots, 7 technical indicators, 5 signal dynamics, 5 side-aware interactions, 5 cross-asset context, and 3 Williams pivot and distress features. Each feature is computed end-of-day and applied with a one-day causal shift.

Table 1: Feature manifest.

Family	Count	Source
Regime (DFT $\rightarrow$ CVAE $\rightarrow$ GMM $\rightarrow$ HMM)	20	Madmoun (2022) Ch. 3
TF-C self-supervised embeddings	32	Zhang et al. (2022) , NeurIPS 2022
Identity one-hots (instrument + asset class)	14	Standard
Technical indicators	7	RSI, ATR, EWMA vol, momentum, Bollinger
Signal dynamics	5	Engineered
Side-aware interactions	5	Engineered
Cross-asset context	5	Engineered
Williams pivots + distress indicator	3	Williams (1979) + practitioner design
Total	91	

The TF-C encoder ([Zhang et al., 2022](#)) is pre-trained on 78,587 length-64 log-return windows from all eleven instruments up to the training cutoff, a corpus spanning the 2000 dot-com crash, the 2008 GFC, the 2018 selloff, and the COVID-19 shock. The

architecture pairs a 1D convolutional time-domain branch with a frequency-domain branch, trained by symmetric NT-Xent contrastive loss in a shared 32-dimensional embedding space. The result is 32-dimensional representations from 24 years of unlabelled history, unavailable to any supervised model restricted to the 2020–2021 signal window.

2.4 Triple-barrier labelling

The triple-barrier label $y_{i,t} \in \{0, 1\}$ (López de Prado, 2018) marks an event as profitable ($y = 1$) if the side-adjusted return path hits the profit-take barrier before the stop-loss within a 10-day vertical window. Four parallel sigma estimators (EWMA symmetric, regime-conditional, asymmetric per-direction, and spectral energy) are computed in parallel; cross-level label agreement ranges from 87% to 96%, confirming robustness to the volatility-scaling choice. Per-event sample uniqueness yields $\bar{u} = 0.018$, giving an effective sample size of $4,968 \times 0.018 \approx 90$.

3 Model Families and Evaluation Protocol

Three primary model families are trained on the same 91-feature panel and the same canonical EWMA label, under an identical evaluation protocol: 200-trial TPE search (Bergstra et al., 2011) on purged 5-fold cross-validation with a 5-day embargo (López de Prado, 2018), per-event uniqueness sample weights, and 30-seed neural-network ensembles for initialisation uncertainty.

Logistic elastic-net. The linear family. TPE selects near-pure lasso behaviour (ℓ_1 -ratio = 0.94), sparsifying the 91-feature panel.

XGBoost. The tree-based family (Chen and Guestrin, 2016). TPE selects `max_depth` = 7 and `n_estimators` = 384, near the upper edge of the tested grid.

GRU-with-attention. The neural family, adapted from Madmoun (2022) Chapter 4. The GRU encoder and Bahdanau attention layer (Bahdanau et al., 2015) are retained while the ranking decoder is removed; the attention-weighted context vector is passed through a sigmoid head. A feed-forward MLP baseline with 2,977 parameters (14% of the GRU+attention’s 21,281) matches it on test within 0.003 AUC, indicating that test performance is bounded by signal rather than by capacity.

Equal-weight ensemble. The deliverable is $\hat{p} = \frac{1}{3}(\hat{p}_{\text{logistic}} + \hat{p}_{\text{XGBoost}} + \hat{p}_{\text{GRU}})$. Equal weighting is deliberate: the forecast-combination literature documents that estimated optimal weights routinely fail to beat the simple average under regime shift, and the effect is strongest with roughly 90 effective observations (Bates and Granger, 1969; Claeskens et al., 2016; Stock and Watson, 2004).

3.1 Results

Table 2: Model-family AUC comparison. With roughly 90 effective observations the ranking is descriptive, not inferential.

Family	Val AUC	Test AUC	Params
Logistic (elastic-net)	0.545	0.482	91
XGBoost	0.503	0.564	n/a
GRU+attention (30-seed)	0.603	0.489	21,281
MLP baseline (10-seed)	0.576	0.486	2,977
Ensemble (deliverable)	0.530	0.509	–

XGBoost is strongest on the visible test (AUC 0.564 [0.507, 0.624] under 5,000-path Politis-Romano bootstrap) but carries the lowest validation AUC and can only be identified as strongest by consulting the held-out set. GRU+attention leads on validation (0.603) but does not transfer to test (0.489); the 30-seed mean is 0.480 with standard deviation 0.022. The ensemble achieves test AUC 0.509 [0.454, 0.567], not statistically separated from random.

4 Feature Importance and the Val-Test Inversion

Cluster-level grouped permutation importance (family cluster-MDA) following [Atluri \(2026a\)](#) identifies TF-C as the dominant contributor across all three model families on validation: +0.037 on the logistic, +0.037 on XGBoost, +0.076 on GRU+attention. Pre-training on 78,587 historical windows up to the training cutoff produces representations no other family matches.

On test, TF-C dominance shrinks by 70% on XGBoost and 84% on GRU+attention. Side-aware interaction features show a striking inversion: on validation, permuting them *improves* XGBoost AUC by 0.035 (they are harmful); on test, the same permutation *drops* AUC by 0.039 (they are helpful). The features encode directional structure that hurts during the training regime but helps during the H1 2022 regime break. A second train-test boundary (Split B: train on 2020 only, validate on H1 2021, test on H2 2021) reproduces the side-aware sign flip with the same H1/H2 directional structure, suggesting a recurring half-year instability rather than a single-split artefact.

5 The H1 2022 Equity Inversion

During training (2020-H1 to 2021-H1), equity index futures rallied strongly, producing 75% profitable long signals and 28% profitable shorts. During the H1 2022 equity draw-

down, longs become 31% profitable and shorts 81% profitable. The learned preference is the wrong preference; equity-index test AUC falls below 0.5 as a direct consequence. This is primarily concept drift in the sense of [Gama et al. \(2014\)](#): the conditional $P(Y | X)$ changes between train and test rather than $P(X)$ or $P(Y)$. Three independent verifications support this reading: the direct profitability inversion; the per-asset-class breakdown localising failure to equity, with energy retaining ranking skill and metals mixed; and a development ablation showing that dropping the three direction-revealing signal features lifts equity test AUC from 0.286 to 0.513, isolating the inversion as the dominant driver rather than feature quality or model capacity. Per-asset-class test AUC confirms: equity 0.486, energy 0.557, metals 0.480. Multi-asset pooling is the most consequential design decision, since equity-only models collapse (AUC 0.286 to 0.332) while pooled training recovers the deliverable ensemble to 0.509.

6 WGAN-GP Synthetic Null

A WGAN-GP ([Wiese et al., 2020](#)) trained on pre-2020 standardised log returns across all eleven instruments serves as a direct overfitting check. The generator reproduces canonical return stylised facts: heavy-tailed kurtosis (synthetic 12.0 vs. real 11.2), negative skew, unit-scale dispersion, and volatility clustering. For each of 200 synthetic draws, H1 2022 price paths are replaced with generator-sampled, drift-removed paths scaled to real EWMA volatility, triple-barrier labels are recomputed, and the already-trained metamodel is scored against them.

XGBoost’s test AUC of 0.564 lies above the null 95% band with $p = 0.015$; its visible-test ranking is unlikely under the signal-free synthetic null. The deliverable ensemble’s test AUC of 0.509 sits inside the null ($p = 0.605$, below the null mean of 0.519). The synthetic null sharpens but does not change the deliverable decision: XGBoost carries the lowest validation AUC and cannot be selected without consulting the test set.

7 Strategy Construction

The strategy track converts ensemble probabilities into sign-preserving, volatility-capped position weights under a 10% annualised volatility constraint. Five tiers are constructed against this constraint, with Tier 0 retained as a raw diagnostic benchmark.

Table 3: Strategy backtest, H1 2022. All metrics gross of transaction costs. Vol cap 10%, soft target 9.5%.

Metric	Tier 0	Tier 1	Tier 1b	Tier 2	Tier 3
	<i>raw</i>	<i>filter</i>	<i>sizer</i>	<i>LW MV</i>	<i>boot MV</i>
H1 Total Return	+26.13%	+3.12%	+3.17%	+10.54%	+10.14%
CAGR	+58.50%	+6.29%	+6.38%	+21.99%	+21.13%
Annualised Vol.	10.09%	8.78%	7.73%	9.48%	9.51%
Sharpe Ratio	4.62	0.74	0.84	2.15	2.06
Sortino Ratio	8.74	1.24	1.32	3.52	3.42
Max Drawdown	−2.33%	−9.25%	−7.61%	−6.78%	−6.81%
Ann. Turnover	120.80	127.35	85.91	211.62	210.41

The unfiltered primary signal (Tier 0) achieves Sharpe 4.62 with maximum drawdown −2.33%; the metamodel filter drops Sharpe to 0.74 (Tier 1) and increases the drawdown to −9.25%. The filter removes trades that were profitable during the equity drawdown because the metamodel, trained on the 2020–2021 long-bias regime, was less confident in those positions and filtered them out. Within the filtered universe, covariance-aware sizing accounts for the entire Sharpe progression from 0.74 to 2.06, exploiting the low correlation between energy, metals, and equity futures under the volatility cap. Transaction-cost sensitivity is the most consequential caveat. At 1 bp per turnover unit, net Sharpe drops from 2.06 to 1.84; at 5 bps, to 0.97; at 10 bps, the strategy loses money.

8 Conclusion

The central finding is that supervised meta-labelling on a primary directional signal does not survive the H1 2022 inversion in the conditional direction-profitability mapping for equity index futures. The cluster-MDA side-aware sign flip reproduces at a second train-test boundary, identifying the inversion as a structural regime-shift signature. Each of the flanking robustness channels is too tight to account for the validation-to-test gap: the XGBoost bootstrap lower bound is 0.507, the GRU initialisation standard deviation is 0.022, label-construction agreement is 87% to 96%, and the synthetic null separates XGBoost from signal-free paths but not the deliverable ensemble. Under the resulting approximately 90 effective independent observations, the equal-weight ensemble is the assumption-minimising deliverable: XGBoost’s stronger visible-test result can only be identified by consulting the held-out set, which is the disqualifying condition for any robustness protocol rather than the basis for selecting the model.

Four implementation limitations qualify these results. First, the GRU+attention cannot be cleanly distinguished from general neural capacity. Second, the 10-day vertical barrier is grounded in labelling consistency and signal-duration matching, not in AUC optimisation. Third, validation cluster-MDA does not predict test cluster-MDA for direction-revealing features under regime shift. Fourth, strategy-level results are gross of transaction costs, and the cost-sensitivity table makes clear that the headline Sharpe does not survive realistic execution costs.

In a live trading setting, the inversion diagnosis would motivate an event-driven architecture conditioned on regime state and dislocation structure rather than a refined metamodel on the same primary signal.

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